Janki Bhimani

Rising Scholar Associate Professor (with tenure)
Knight Foundation School of Computing and Information Science
Florida International University, Miami, Florida
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RESEARCH PROFILE

Dr. Janki Bhimani is a tenured Rising Scholar Associate Professor in the Knight Foundation School of Computing and Information Science at Florida International University. Her research advances intelligent, efficient, and resilient computing systems by addressing fundamental challenges in memory and storage, AI infrastructure, and emerging quantum-classical systems.

She has established an externally funded research program supported by more than $6.5M in competitive awards, including an NSF CAREER Award and more than $3.0M in awards as Principal Investigator. Her record includes 61 publications, more than 1,300 citations, 12 patent disclosures, applications, and/or granted patents (see patent section for status), and multiple best-paper recognitions. Her publications span systems, electronic design automation, and computing venues including OSDI, FAST, EuroSys, HPCA, IPDPS, ICCAD, DAC, and AISTATS, as well as ACM and IEEE journals.

Her research agenda connects three directions: (1) intelligent memory, storage, and AI systems; (2) reliable and adaptive computing, including operational intelligence and quantum-classical system resilience; and (3) AI-enabled applications, including neuro-symbolic agents for undergraduate STEM advising. Across these directions, her work integrates system design, empirical evaluation, and practical research artifacts to address challenges spanning hardware, software, and intelligent decision-making.

She is also an educator, doctoral advisor, and academic leader. She has developed and taught courses in storage systems, data structures, and quantum computing; mentored Ph.D., M.S., undergraduate, and REU researchers; and contributed to journal editorship, conference leadership, NSF review, and university governance.

RESEARCH AREAS

Core Area: Computer Systems and Architecture
Quantum Computing and Quantum Systems: Quantum-classical systems, quantum algorithm execution and optimization, quantum systems resiliency, fault tolerance, and AI-assisted quantum computing.
Memory, Storage, and Data Systems: Flash and NVMe storage, emerging memory technologies, CXL and disaggregated memory, AI data pipelines, caching, checkpointing, and resource management.
AI for Systems and Intelligent Computing: Machine learning for systems optimization, AI-driven resource management, operational intelligence, and reliable computing.
AI-Enabled Applications: Neuro-symbolic AI, intelligent agents, and AI-assisted STEM education.

ACADEMIC & INDUSTRY APPOINTMENTS

Rising Scholar Associate Professor with Tenure Jun 2025 – Present
Florida International University *Miami, FL*

Visiting Faculty / Sabbatical Researcher Spring 2026
BITS Pilani *Goa, India*

Assistant Professor (Tenure-Track) Aug 2019 – Jun 2025
Florida International University *Miami, FL*

Affiliated Faculty Aug 2019 – Jul 2025
Center for Women's and Gender Studies, FIU *Miami, FL*

Performance Architect Intern May 2016 – Aug 2018 (Summers)
Samsung Semiconductors Inc. Research Lab *San Jose, CA*

Graduate Research & Teaching Assistant May 2014 – Jun 2019
Northeastern University *Boston, MA*

Education

Ph.D. in Computer Engineering Aug 2019
Northeastern University *Boston, MA*

- *Dissertation:* Enhancing Efficiency and Endurance of Flash-Based Storage for Big Data Processing on Enterprise Cloud and Datacenter

M.S. in Computer Engineering Jan 2016
Northeastern University *Boston, MA*

- *M.S. research:* FiM - Fine grained Model to Predict Heterogeneous Computing Platforms Performance

B.Tech. in Electrical & Electronics Engineering Aug 2013
GITAM University *India*

- *Major:* Robotics and Programming of Embedded Systems, Minor: Circuit Design, Power Management

PROFESSIONAL ACHIEVEMENTS, HONORS, AWARDS, AND FELLOWSHIPS

(Content in blue color are items since arriving at FIU.)

1. **ASEMFL Rising Star in Science Award** 2026
2. **ACM SAC Best Paper Award**, 41st ACM/SIGAPP Symposium on Applied Computing (SAC '26) 2026
3. **Distinguished Reviewer Award**, 13th ACM Workshop on Hot Topics in Storage and File Systems (HotStorage '25) 2025
4. **IEEE Senior Member Grade**, Institute of Electrical and Electronics Engineers (IEEE) 2025
5. **Rising Scholar Award**, Florida International University (FIU) 2025
6. **In the Company of Women Award**, Miami-Dade County 2025
7. **NSF CAREER Award**, National Science Foundation 2024
8. **FIU Top Scholar Award**, Category: Research, Creative Activities, & Award-Winning Publications 2023
9. **Quality Matters Certification**, Online Course: CIS5346 Storage System 2023
10. **Outstanding Applied Research Award**, Knight Foundation School of Computing and Information Science (KFSCIS), FIU 2022
11. **ACM HotStorage Best Paper Award**, 14th ACM Workshop on Hot Topics in Storage and File Systems (HotStorage '22) 2022
12. **DATE Best Paper Award Nomination**, Design, Automation and Test in Europe Conference (DATE '22) 2022
13. **Quality Matters Certification**, Online Course: COP3530 Data Structures 2022
14. **Certificate of Completion**, ASEE DELTA Junior Faculty Institute 2021
15. **Grace Hopper Celebration (GHC) Faculty Scholarship**, Women in Computing 2021
16. **Distinguished Reviewer Award**, 13th ACM Workshop on Hot Topics in Storage and File Systems (HotStorage '21) 2021

17. **Hybrid Course Certification**, COP3530 Data Structures 2020
18. **Grace Hopper Celebration (GHC) Faculty Scholarship**, Women in Computing 2020
19. **Outstanding Graduate Research Award**, Northeastern University 2019
20. **IEEE CLOUD Best Paper Award**, 11th IEEE International Conference on Cloud Computing 2018
21. **IEEE IPCCC Best Paper Award**, 36th IEEE International Performance Computing and Communications Conference 2017
22. **Double Husky Scholarship**, Northeastern University 2014
23. **Best Budget Robot Award**, 3rd Lunabotics International Mining Competition, NASA Kennedy Space Center, FL 2012
24. **Best Working Model Award**, Junk Yard Wars, Conscientia, Indian Institute of Space Science Technology (IIST) 2012
25. **Best Paper Award**, Aagama National Level Technical Paper Contest 2012
26. **Best Working Model Award**, Junk Yard Wars, Technozion at National Institute of Technology (NIT) 2011
27. **Outstanding Debate Performance Award**, Institute of Engineers India (IEI) 2011
28. **Impromptu Speaker Award**, International Society for Technology in Education (ISTE) 2010
29. **University Merit Scholarship**, GITAM University 2010–2013

FUNDED RESEARCH GRANTS

[14 PI + 4 Co-PI/SI = $6,502,201; Total PI funds = $3,002,232; My share of PI funds to FIU = $1,982,303; Co-PI/SI funds to FIU = $215,745.79]

1. **2026 - 2029 NSF IUSE (Co-PI)**
   "DUE-2613794: Artificial Intelligence-Assisted Socially Interactive Agents to Transform Undergraduate STEM Advising"
   PI team: Agoritsa Polyzou, Florida International University; Christine Lisetti, Florida International University; Tiberio Garza, Florida International University (Senior Personnel)
   Total Value: $399,983 (Direct+Indirect) My share: $119,995 (30%)
   Start date: Oct 15, 2026 Expiration date: Sep 30, 2029
   Project ID: 800023121
2. **2026-2027 NSF REU Supplement (Only PI)**
   "CAREER-2338457 - Towards Efficient In-storage Indexing"
   Total Value: $10,000 (Direct+Indirect) My share: $10,000 (100%)
   Start date: Jul 1, 2026 Expiration date: Jun 30, 2029
   Project ID: CRA Managed
3. **2026-2027 DOE LBNL NERSC QCAN IBM Quantum Access Grant (Only PI)**
   "ERCAP0035598: Towards Resiliency of the Quantum Systems"
   Total Value: 30min/month Quantum Access My share: 100%
   Start date: Feb 1, 2026 Expiration date: Aug 31, 2026
4. **2025-2026 DOE NERSC Perlmutter Supercomputer Access (Only PI)**
   "GPU-CPU Access: Designing Frameworks for GPU Virtualization"
   Total Value: 9000 hrs of CPU and 6000 hrs of GPU Access My share: 100%
   Start date: Aug 1, 2025 Expiration date: Jul 31, 2026
5. **2025-2026 DOE LBNL NERSC QCAN IBM Quantum Access Grant (Only PI)**
   "ERCAP0034716: Quantum Checkpointing"
   Total Value: 120min/month Quantum Access My share: 100%

Start date: Aug 1, 2025 Expiration date: Jan 31, 2026

6. **2024 - 2028 NSF Core Medium (PI)**
"CSR-2402328: Medium: DISCO: Disciplined Data Science Framework for Storage I/O Management"
PI team: Haryadi S. Gunawi, University of Chicago and Sandeep R. Madireddy, Argonne National Laboratory
Total Value: $1,200,000 (Direct+Indirect) My share: $475,000 (37.5%)
Start date: Oct 1, 2024 Expiration date: Sep 30, 2028
Project ID: 800019285
7. **2024-2029 NSF CAREER (Only PI)**
"CAREER-2338457 - Towards Efficient In-storage Indexing"
Total Value: $615,528 (Direct+Indirect) My share: $615,528 (100%)
Start date: Jul 1, 2024 Expiration date: Jun 30, 2029
Project ID: 800018670
8. **2024-2026 NSF REU Supplement (Only PI)**
"CSR-2406069 - REU: Learning and Management in Tiered Memory Systems"
Total Value: $16,000 (Direct+Indirect) My share: $16,000 (100%)
Start date: Jan 1, 2024 Expiration date: Dec 31, 2024
Project ID: 800020019
9. **2023-2026 NSF CISE Core (Only PI)**
"CSR-2323100 NSF Core: CSR: Small: Learning and Management in Tiered Memory Systems"
Total Value: $514,704 (Direct+Indirect) My share: $514,704 (100%)
Start date: Oct 1, 2023 Expiration date: Sep 30, 2026
Project ID: 800018181
10. **2023-2024 Samsung Global Research Outreach (GRO) Award (PI)**
"Leveraging Disaggregated Servers for Large Scale AI Training Acceleration"
Co-PI: Bryan Kim, Syracuse University
Total Value: $50,000 (Direct+Indirect) My share: $25,000 (50%)
Start date: Sep 1, 2023 Expiration date: Feb 29, 2024

11. **2022-2027 NSF HSI (SI)**
"HRD-2225201 - HSI Institutional Transformation Project Voces (Voices for Organizing Change in Educational Systems)"
PI team: Yesim Darici, Stephen Secules, Rocio Benabentos, Laird Kramer, Jaroslava Miksovska, Monica Cardella, FIU
Total Value: $2,999,986 (Direct+Indirect) My share: $69,340 (2.3%)
Start date: Sep 15, 2022 Expiration date: Aug 31, 2027

12. **2022-2023 Samsung Memory Solutions Lab (MSL) Research Award (PI)**
"Leveraging Disaggregated Servers for Large Scale AI Training Acceleration"
Co-PI: Bryan Kim, Syracuse University
Total Value: $50,000 (Direct+Indirect) My share: $25,000 (50%)
Start date: Mar 1, 2023 Expiration date: Aug 31, 2023

13. **2021 Samsung Semiconductor Inc. Equipment Grant (Only PI)**
"Parallel Data Access with Key-Value SSDs"
Total Value: $10,000 (Direct+Indirect) My share: $10,000 (100%)
Start date: Oct 1, 2021 Expiration date: Jun 30, 2022
14. **2021-2022 NSF REU Supplement (Only PI)**

"CNS-2122987 - REU: New Techniques for I/O Behavior Modeling and Persistent Storage Device Configuration"
Total Value: $16,000 (Direct+Indirect) My share: $16,000 (100%)
Start date: May 1, 2021 Expiration date: Apr 30, 2022
Project ID: 800014726

15. **2020-2023 NSF CISE Core (Lead PI)**
"CNS-2008324 - Small: New Techniques for I/O Behavior Modeling and Persistent Storage Device Configuration"
Co-PI: Ningfang Mi, Northeastern University
Total Value: $500,000 (Direct+Indirect) My share: $255,071 (51%)
Start date: May 1, 2020 Expiration date: Apr 30, 2023
Project ID: 800012359

16. **2020-2022 Cyber Florida (Co-PI)**
"RumorHunt: A Next-Generation Online Scalable Streaming System"
PI team: Liting Hu, FIU and Zhishan Guo, University of Central Florida
Total Value: $75,000 (Direct+Indirect) My share: $21,410.79 (28%)
Start date: Aug 1, 2020 Expiration date: May 30, 2022
Project ID: 800012574

17. **2019-2020 FIU Faculty Grantsmanship Development Program (Co-PI)**
"Design, Development and Testing of Distributed Computing Framework for globally coordinated data submission and accessibility of Mass Spectrometry Data"
PI team: Fahad Saeed, Alex Afanasyev, Hadi Amini, FIU
Total Value: $25,000 (Direct+Indirect) My share: $5,000 (20%)
Start date: Nov 1, 2019 Expiration date: May 30, 2020

18. **2019 Samsung Semiconductor Inc. Equipment Grant (Only PI)**
"Exploring Vulnerabilities of Key-Value SSDs"
Total Value: $20,000 (Direct+Indirect) My share: $20,000 (100%)
Start date: Oct 1, 2019 Expiration date: Sep 30, 2021

PUBLICATIONS IN DISCIPLINE

Papers in blue color are publications at FIU. *(* Indicates an FIU student supervised by myself.)*
Citation counts are taken from my Google Scholar Profile, which lists the following statistics: Total citations: 1300, h-index: 19, i10-index: 31.

[Total 61 - 36 FIU (with FIU students supervised by myself) and 25 Pre-FIU]

Selective Refereed Journal Publications

1. Manoj P. Saha, Yanzhao Wu, Cláudia Brito, Raju Rangaswami, João Paulo, Ricardo Macedo, Janki Bhimani. Fragment: Efficient DNN Checkpoint with Relaxed Model Consistency. ACM Transactions on Architecture and Code Optimization (TACO), 2026. Tier 1 Journal with impact factor 2.3 **(CORE2020 B; Google Scholar Metrics)**
2. Xiangqun Zhang, Janki Bhimani, Shuyi Pei, Eunji Lee, Sungjin Lee, Yoon Jae Seong, Eui Jin Kim, Changho Choi, Eyee Hyun Nam, Jongmoo Choi, Bryan S. Kim. Storage Abstractions for SSDs: The Past, Present, and Future. ACM Transactions on Storage (TOS) 2024. Tier 1 Journal with impact factor 2.1 **(CORE2020 B; Google Scholar Metrics)**
3. Danlin Jia, Li Wang, Natalia Valencia*, Janki Bhimani, Bo Sheng and Ningfang Mi. Learning-based Dynamic Memory Allocation Schemes for Apache Spark Data Processing. IEEE Transactions on Cloud Computing (TCC) 2023. Tier 1 Journal with impact factor 11.1 **(Google Scholar Metrics)**
4. Ajinkya S Bankar, Shi Sha, Janki Bhimani, Vivek Chaturvedi, Gang Quan. Thermal Aware System-

Wide Reliability Optimization for Automotive Distributed Computing Applications. IEEE Transactions on Vehicular Technology (TVT) 2022. Tier 1 Journal with impact factor 2.243 **(Google Scholar Metrics)**

5. Janki Bhimani, Zhengyu Yang, Jingpei Yang, Adnan Maruf*, Ningfang Mi, Rajinikanth Pandurangan, Changho Choi, Vijay Balakrishnan. Automatic Stream Identification to Improve Flash Endurance in Data Centers. ACM Transactions on Storage (TOS) 2021. Tier 1 Journal with impact factor 1.59 **(CORE2020 B; Google Scholar Metrics)**
6. Janki Bhimani, Adnan Maruf*, Ningfang Mi, Rajinikanth Pandurangan, and Vijay Balakrishnan. Auto-Tuning Parameters for Emerging Multi-Stream Flash-Based Storage Drives Through New I/O Pattern Generations. IEEE Transactions on Computers (TC) 2020. Tier 1 Journal with impact factor 3.131 **(CORE2020 A*; Google Scholar Metrics)**
7. Janki Bhimani, Ningfang Mi, Miriam Leeser, and Zhengyu Yang, New Performance Modeling Methods for Parallel Data Processing Applications, ACM Transactions on Modeling and Computer Simulation (TOMACS), 2019. DOI 10.1145/3309684. Tier 1 Journal with impact factor 1.380 **(CORE2020 B; Google Scholar Metrics)**
8. Zhengyu Yang, Manu Awasthi, Mrinmoy Ghosh, Janki Bhimani, and Ningfang Mi, I/O Workload Management for All-Flash Datacenter Storage Systems Based on Total Cost of Ownership, IEEE Transactions on Big Data (TBDATA), Special Issue on the Integration of Extreme Scale Computing and Big Data Management and Analytics, 2018. DOI 10.1109/TBDATA.2018.2871114. Tier 1 Journal with impact factor 2. **(Google Scholar Metrics)**
9. Janki Bhimani, Zhengyu Yang, Ningfang Mi, Jingpei Yang, Qiumin Xu, Manu Awasthi, Rajinikanth Pandurangan, and Vijay Balakrishnan, Docker Container Scheduler for I/O Intensive Applications running on NVMe SSDs, IEEE Transactions on Multi-Scale Computing Systems (TMSCS), 2018. DOI: 10.1109/TMSCS.2018.2801281. Tier 1 Journal with impact factor 2.065. **(Google Scholar Metrics)**
10. Zhengyu Yang, Janki Bhimani, Yi Yao, Cho-Hsien Lin, Jiayin Wang, Ningfang Mi, and Bo Sheng, AutoAdmin: Admission Control in YARN Clusters Based on Dynamic Resource Reservation, Scalable Computing: Practice and Experience, Special Issue on Advances in Emerging Wireless Communications and Networking (SCPE), 2018. Volume 19, Number 1, pp. 53–67.
11. Zhengyu Yang, Yufeng Wang, Janki Bhimani, Chiu C. Tan, and Ningfang Mi, EAD: Elasticity Aware Deduplication Manager for Datacenters with Multi-tier Storage Systems, Cluster Computing (CC), 2018. https://doi.org/10.1007/s10586-018-2141-z. **(Google Scholar Metrics)**
12. Zhengyu Yang, Janki Bhimani, Jiayin Wang, David Evans, and Ningfang Mi, Automatic and Scalable Data Replication Manager in Distributed Computation and Storage Infrastructure of Cyber-Physical Systems, Scalable Computing: Practice and Experience, Special Issue on Communication, Computing, and Networking in Cyber-Physical Systems (SCPE), 2018. Volume 18, Number 4, pp. 291–311.

Highly Selective Peer Reviewed Conference Publications
Acceptance rates below 30%

13. Mayur Akewar, Dongsheng Luo, Sandeep Madireddy, Janki Bhimani, SMARTTalk: Teaching SMART Logs to Talk to LLMs, 20th USENIX Symposium on Operating Systems Design and Implementation (OSDI), Seattle, WA, USA, July 13–15, 2026. **(Acceptance Rate: 17.2%; ICORE A*; CSRankings-selected; Google Scholar Metrics)**
14. Christopher Kverne, Mayur Akewar, Yuqian Huo, Tirthak Patel, Janki Bhimani, WSBD: Freezing-Based Optimizer for Quantum Neural Networks, 2026 International Conference on Artificial Intelligence and Statistics (AISTATS), Tangier, Morocco. **(Acceptance Rate: 28%; ICORE A; Google Scholar Metrics)**
15. José Pedro Peixoto, Alexis Gonzales*, Janki Bhimani, Raju Rangaswami, Cláudia Brito, João Paulo, and Ricardo Macedo, Holpaca: Holistic and Adaptable Cache Management for Shared Environments,

2026 The International Conference on Performance Engineering (ICPE), Florence, Italy. **(ICORE B)**

16. Mayur Akewar*, Sandeep Madireddy, Dongsheng Luo, and Janki Bhimani, KORAL: Knowledge Graph Guided LLM Reasoning for SSD Operational Analysis, 2026 The 40th IEEE International Parallel & Distributed Processing Symposium (IPDPS), New Orleans, USA. **(Acceptance Rate: 23.3%; ICORE A; Google Scholar Metrics)**
17. Lorena Amanda Quincoso Lugones*, Christopher Kverne*, Nityam Sharadkumar Bhimani, Ana Carolina Oliveira*, Agoritsa Polyzou, Christine Lisetti, and Janki Bhimani, Aurora: Neuro-Symbolic AI Driven Advising Agent, 2026 The 41st ACM/SIGAPP Symposium On Applied Computing (ACM SAC'26), Thessaloniki, Greece. **(Best Paper Award) (Acceptance Rate: 23.4%; ICORE Multiconference)**
18. Omkar Desai, Ziyang Jiao*, Shuyi Pei, Janki Bhimani, Bryan S. Kim, Preparation Meets Opportunity: Enhancing Data Preprocessing for ML Training With Seneca, 2026 24th USENIX Conference on File and Storage Technologies (FAST'26), Santa Clara. **(Acceptance Rate: 17.4%; ICORE A; CSRankings-selected; Google Scholar Metrics)**
19. Manoj Pravakar Saha*, Ashikee Ghosh*, Raju Rangaswami, Yanzhao Wu, and Janki Bhimani, LATTICE: Efficient In-Memory DNN Model Versioning, 2025 ACM International Systems and Storage Conference (SYSTOR'25), Virtual. **(Acceptance Rate: 37.2%; ICORE National: Israel)**
20. Yuqian Huo, Jinbiao Wei, Christopher Kverne*, Mayur Akewar*, Janki Bhimani, and Tirthak Patel, Revisiting Noise-adaptive Transpilation in Quantum Computing: How Much Impact Does it Have?, 2025 IEEE/ACM International Conference on Computer-Aided Design (ICCAD'25), Munich, Germany. **(Acceptance Rate: 24.7%; ICORE A; CSRankings-selected; Google Scholar Metrics)**
21. Christopher Lukas Kverne*, Mayur Akewar*, Yuqian Huo, Tirthak Patel, and Janki Bhimani, Quantum Neural Networks Need Checkpointing, 2025 ACM Workshop on Hot Topics in Storage and File Systems (HotStorage '25), Boston. **(Acceptance Rate: 33%; ICORE Not Ranked)**
22. Mayur Akewar*, Gang Quan, Sandeep Madireddy, and Janki Bhimani, Can LLMs Model the Environmental Impact on SSD?, 2025 ACM Workshop on Hot Topics in Storage and File Systems (HotStorage '25), Boston. **(Acceptance Rate: 33%; ICORE Not Ranked)**
23. Christopher Lukas Kverne*, Federico Monteverdi*, Agoritsa Polyzou, Christine Lisetti, and Janki Bhimani, Course-Job Fit: Understanding the Contextual Relationship Between Computing Courses and Employment Opportunities, 2025 ASEE Annual Conference (ASEE'25), Montreal, QC, Canada. **(Acceptance Rate: 20%; ERA2010 B)**
24. Daniar Kurniawan, Maharani A. Putri, Peiran Qin, Kahfi S. Zulkifli, Ray A. O. Sinurat, Janki Bhimani, Sandeep Madireddy, Achmad Imam Kistijantoro, and Haryadi Gunawi, Heimdall: Optimizing Storage I/O Admission with Extensive Machine Learning Pipeline, 2025 EuroSys ACM SIGOPS European chapter (EuroSys'25), Rotterdam. **(Acceptance Rate: 12.2%; ICORE A; CSRankings-selected; Google Scholar Metrics)**
25. Manoj Saha*, Danlin Jia, Janki Bhimani and Ningfang Mi, MoKE: Modular Key-value Emulator for Realistic Studies on Emerging Storage Devices, 2023 IEEE International Conference on Cloud Computing (CLOUD'23), Hybrid Event, Chicago, IL, 2023. **(ICORE B; Google Scholar Metrics)**
26. Ziyang Jiao, Janki Bhimani, Bryan S. Kim, Wear Leveling in SSDs Considered Harmful, 2022 ACM Workshop on Hot Topics in Storage and File Systems (HotStorage '22), Virtual. **(Best Paper Award)**
27. Adnan Maruf*, Sashri Brahmakshatriya*, Baolin Li, Devesh Tiwari, Gang Quan and Janki Bhimani, Do Temperature and Humidity Exposures Hurt or Benefit Your SSDs?, 2022 Design, Automation and Test in Europe Conference. The European Event for Electronic System Design and Test (DATE'22), Virtual. **(Acceptance Rate: 25%; Best Paper Award Nomination) (ICORE A; Google Scholar Metrics)**
28. Adnan Maruf*, Ashikee Ghosh*, Janki Bhimani, Daniel Campello, Andy Rudoff, Raju Rangaswami,

MULTI-CLOCK: Dynamic Tiering for Hybrid Memory Systems, 2022 IEEE International Symposium on High-Performance Computer Architecture (HPCA'22), Seoul, South Korea, 2022. **(Acceptance Rate: 29.3%; ICORE A*; CSRankings-selected; Google Scholar Metrics)**

29. Adnan Maruf*, Zhengyu Yang, Bridget Davis, Daniel Kim, Jeffrey Wong, Matthew Durand, and Janki Bhimani, Understanding Flash-Based Storage I/O Behavior of Games, 2021 IEEE International Conference on Cloud Computing (CLOUD'21), Online Virtual Congress, 2021. **(IEEE Micro Top Picks from the Computer Architecture Conferences Nomination) (Acceptance Rate: 23.8%; ICORE B; Google Scholar Metrics)**
30. Janki Bhimani, Jingpei Yang, Ningfang Mi, Changho Choi, and Manoj Pravakar Saha*, Fine-grained Control of Concurrency within KV-SSDs, 2021 14th ACM International Systems and Storage Conference (SYSTOR'21), Virtual. **(Acceptance Rate: 29.9%; ICORE National: Israel)**
31. Manoj Pravakar Saha*, Bryan Kim, and Janki Bhimani, KV-SSD: What is it Good For?, 2021 Design Automation Conference (DAC'21), San Francisco, CA, 2021. **(Acceptance Rate: 23%; ICORE A*; CSRankings-selected; Google Scholar Metrics)**
32. Danlin Jia, Manoj Pravakar Saha*, Janki Bhimani, and Ningfang Mi, Performance and Consistency Analysis for Distributed Deep Learning Applications, 2020 International Performance Computing and Communications Conference (IPCCC'20), Virtual using Zoom, 2020. **(Acceptance Rate: 29.3%; ICORE C)**
33. Janki Bhimani, Rajinikanth Pandurangan, Ningfang Mi, and Vijay Balakrishnan, Emulate Processing of Assorted Database Server Applications on Flash-Based Storage in Datacenter Infrastructures, 2019 International Performance Computing and Communications Conference (IPCCC'19), London, UK, 2019. **(Acceptance Rate: 29.2%; ICORE C)**
34. Danlin Jia, Janki Bhimani, Son Nam Nguyen, Bo Sheng, and Ningfang Mi, ATuMm: Auto-tuning Memory Manager in Apache Spark, 2019 International Performance Computing and Communications Conference (IPCCC'19), London, UK, 2019. **(Acceptance Rate: 29.2%; ICORE C)**
35. Janki Bhimani, Tirthak Patel, Ningfang Mi, and Devesh Tiwari, "What does Vibration do to Your SSD?", 2019 Design Automation Conference (DAC'19), Las Vegas, NV, 2019. **(Acceptance Rate: 24.3%; ICORE A*; CSRankings-selected; Google Scholar Metrics)**
36. Janki Bhimani, Ningfang Mi, Zhengyu Yang, Jingpei Yang, Rajinikanth Pandurangan, Changho Choi and Vijay Balakrishnan, "FIOS: Feature Based I/O Stream Identification for Improving Endurance of Multi-Stream SSDs", 2018 IEEE International Conference on Cloud Computing (CLOUD'18), San Francisco, CA, 2018. **(Best Paper Award) (Acceptance Rate: 15%; ICORE B; Google Scholar Metrics)**
37. Janki Bhimani, Ningfang Mi, and Bo Sheng, "BloomStream: Data Temperature Identification for Flash Based Memory Storage Using Bloom Filters", 2018 IEEE International Conference on Cloud Computing (CLOUD'18), San Francisco, CA, 2018. **(Acceptance Rate: 15%; ICORE B; Google Scholar Metrics)**
38. Zhengyu Yang, Morteza Hoseinzadeh, Ping Wong, John Artoux, Clay Mayers, David Thomas Evans, Rory Thomas Bolt, Janki Bhimani, Ningfang Mi, and Steven Swanson, "H-NVMe: A Hybrid Framework of NVMe-based Storage System in Cloud Computing Environment", IEEE International Performance Computing and Communications Conference (IPCCC'17), San Diego, CA, 2017. **(Best Paper Award) (Acceptance Rate: 32.7%; ICORE C)**
39. Zhengyu Yang, Morteza Hoseinzadeh, Allen Andrews, Clay Mayers, David Thomas Evans, Rory Thomas Bolt, Janki Bhimani, Ningfang Mi, and Steven Swanson, "AutoTiering: Automatic Data Placement Manager in Multi-Tier All-Flash Datacenter", IEEE International Performance Computing and Communications Conference (IPCCC'17), San Diego, CA, 2017. **(Acceptance Rate: 32.7%; ICORE C)**

40. Janki Bhimani, Ningfang Mi, Miriam Leeser, and Zhengyu Yang, "FiM: Performance Prediction Model for Parallel Computation in Iterative Data Processing Applications", IEEE International Conference on Cloud Computing (CLOUD'17), Honolulu, HI, 2017. **(Acceptance Rate: 18%; ICORE B; Google Scholar Metrics)**
41. Han Gao, Zhengyu Yang, Janki Bhimani, Teng Wang, Jiayin Wang, Ningfang Mi, and Bo Sheng, "AutoPath: Harnessing Parallel Execution Paths for Efficient Resource Allocation in Multi-Stage Big Data Frameworks", International Conference on Computer Communications and Networks (ICCCN'17), Vancouver, Canada, 2017. **(Acceptance Rate: 25%; ICORE B)**
42. Qiumin Xu, Manu Awasthi, Krishna T. Malladi, Janki Bhimani, Jingpei Yang, and Murali Annavaram. "Performance analysis of containerized applications on local and remote storage" International Conference on Massive Storage Systems and Technology (MSST'17), Santa Clara, CA, 2017. **(Acceptance Rate: 31.6%; ICORE National: USA)**
43. Janki Bhimani, Jingpei Yang, Zhengyu Yang, Ningfang Mi, Qiumin Xu, Manu Awasthi, Rajinikanth Pandurangan, Changho Choi and Vijay Balakrishnan, "Understanding Performance of I/O Intensive Containerized Applications for NVMe SSDs", IEEE International Performance Computing and Communications Conference (IPCCC'16), Las Vegas, NV, 2016. **(Acceptance Rate: 25.5%; ICORE C)**
44. Zhengyu Yang, Jianzhe Tai, Janki Bhimani, Jiayin Wang, Ningfang Mi, and Bo Sheng, "GREM: Dynamic SSD Resource Allocation in Virtualized Storage Systems with Heterogeneous VMs", IEEE International Performance Computing and Communications Conference (IPCCC'16), Las Vegas, NV, 2016. **(Acceptance Rate: 25.50%; ICORE C)**

Other Peer Reviewed Conference and Workshop Publications

Acceptance rates provided when known

45. André Ferreira, Gonçalo Sousa, Amit Ruhela, John Cazes, Janki Bhimani, Raju Rangaswami, Ricardo Macedo, Cláudia Brito and João Paulo, "One Does Not Simply Cache: A Study on Distributed Caches for AI Training in HPC," SRDS 2025 Workshop on Reliable Large-scale Data Management (RLDM'25), Porto, Portugal, 2025. **(Workshop; associated with SRDS, ICORE B)**
46. José Peixoto, Alexis Gonzalez*, Janki Bhimani, Raju Rangaswami, Cláudia Brito, João Paulo and Ricardo Macedo, "Idiosyncrasies of Programmable Caching Engines," SRDS 2025 Workshop on Reliable Large-scale Data Management (RLDM'25), Porto, Portugal, 2025. **(Workshop; associated with SRDS, ICORE B)**
47. Manoj P. Saha*, Omkar Desai, Bryan S. Kim, Janki Bhimani, "Leveraging Keys In Key-Value SSD for Production Workloads," The International ACM Symposium on High-Performance Parallel and Distributed Computing (HPDC'23), Orlando, FL, 2023. (Short Paper) **(Acceptance Rate: 21.2%; ICORE A; CSRankings-selected; Google Scholar Metrics)**
48. Adnan Maruf*, Daniel Carlson*, Ashikee Ghosh*, Manoj Saha*, Janki Bhimani, Raju Rangaswami, "Allocation Policies Matter for Hybrid Memory Systems," The International ACM Symposium on High-Performance Parallel and Distributed Computing (HPDC'23), Orlando, FL, 2023. (Short Paper) **(Acceptance Rate: 21.2%; ICORE A; CSRankings-selected; Google Scholar Metrics)**
49. Manoj P. Saha*, Bryan S. Kim, Haryadi S. Gunawi, Janki Bhimani. "RHIK - Re-configurable Hash-based Indexing for KVSSD," The International ACM Symposium on High-Performance Parallel and Distributed Computing (HPDC'23), Orlando, FL, 2023. (Short Paper) **(Acceptance Rate: 21.2%; ICORE A; CSRankings-selected; Google Scholar Metrics)**
50. Mahsa Bayati, Janki Bhimani, Ronald Lee, Ningfang Mi. "Exploring Benefits of NVMe SSDs for BigData Processing in Enterprise Data Centers," International Conference on Big Data Computing and Communication (BIGCOM'19), Qingdao, China, 2019.
51. Janki Bhimani, Jingpei Yang, Zhengyu Yang, Ningfang Mi, NHV Krishna Giri, Rajinikanth Panduran-

gan, Changho Choi, and Vijay Balakrishnan. "Enhancing SSDs with multi-stream: What? why? how?" IEEE International Performance Computing and Communications Conference (IPCCC'17), San Diego, CA, 2017. (Short Paper) **(ICORE C)**

52. Janki Bhimani, Zhengyu Yang, Miriam Leeser, and Ningfang Mi, "Accelerating Big Data Applications Using Lightweight Virtualization Framework on Enterprise Cloud," IEEE High Performance Extreme Computing Conference (HPEC'17), Waltham, MA, 2017.
53. Qiumin Xu, Manu Awasthi, Krishna T. Malladi, Janki Bhimani, Jingpei Yang, Murali Annavaram, "Docker Characterization on High Performance SSDs," IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS'17), Santa Rosa, CA, 2017. (Short Paper) **(ICORE B)**
54. Liu Chao, Janki Bhimani, and Miriam Leeser, "Using High Level GPU Tasks to Explore Memory and Communications Options on Heterogeneous Platforms," ACM Workshop on Software Engineering Methods for Parallel and High Performance Applications (SEM4HPC), Washington, D.C., 2017.
55. Liu Chao, Janki Bhimani, and Miriam Leeser, "Exploring Memory Options for Data Transfer on Heterogeneous Platforms," The International ACM Symposium on High-Performance Parallel and Distributed Computing (HPDC'17), Washington, D.C., 2017. (Short Paper) **(Acceptance Rate: 19%; ICORE A; CSRankings-selected; Google Scholar Metrics)**
56. Janki Bhimani, Miriam Leeser, and Ningfang Mi, "Performance Prediction Techniques for Scalable Large Data Processing in Distributed MPI Systems," IEEE International Performance Computing and Communications Conference (IPCCC'16), Las Vegas, NV, 2016. Acceptance Rate: 12%. (Short Paper) **(ICORE C)**
57. Janki Bhimani, Miriam Leeser, and Ningfang Mi, "Design Space Exploration of GPU Accelerated Cluster Systems for Optimal Data Transfer Using PCIe Bus," IEEE High Performance Extreme Computing Conference (HPEC'16), Waltham, MA, 2016.
58. Janki Bhimani, Miriam Leeser, and Ningfang Mi, "Accelerating K-Means Clustering with Parallel Implementations and GPU Computing," IEEE High Performance Extreme Computing Conference (HPEC'15), Waltham, MA, 2015.
59. Janki Bhimani, Miriam Leeser and Ningfang Mi, "Predicting the Performance of Machine Learning Algorithms running on Heterogeneous Computing Platforms," Women in Machine Learning Workshop (WiML'14), Montréal, Canada, 2014. (Short Paper)
60. Baiyu Chen, Zhengyu Yang, Siyu Huang, Xianzhi Du, Zhiwei Cui, Janki Bhimani, Xin Xie, and Ningfang Mi, "Cyber-Physical System Enabled Nearby Traffic Flow Modelling for Autonomous Vehicles," IEEE International Performance Computing and Communications Conference, Special Session on Cyber Physical Systems: Security, Computing, and Performance (IPCCC CPS'17), San Diego, CA, 2017. **(ICORE C)**
61. Xianfei Xia, Hongru Xiao, Zhengyu Yang, Xin Xie, and Janki Bhimani, "Pelletization Characteristics of the Hydrothermal Pretreated Rice Straw with Added Binders," Arabian Journal for Science and Engineering 43, no. 9 (2018): 4811–4820.

Books N/A
Chapters in Books N/A
Government Reports or Monographs N/A
Book Reviews

1. **Book Proposal Reviewer, CRC Press–Taylor & Francis, 2026.**
2. Introduction to Data Storage Systems and Services by Amit Golander and Assaf Natanzon. Provided a detailed assessment of the manuscript's pedagogical organization, storage systems framework, technical scope, and relevance to undergraduate and graduate systems education, with recommendations on technical depth, system design trade-offs, abstraction layers, and emerging AI storage workloads.

PRESENTED PAPERS, AND LECTURES

Talks in blue color are given after joining FIU.

1. Guest Speaker: Advancing System Design: Classical and Quantum, IIT Gandhinagar, March 13, 2026.
2. Invited Expert Lecture: Flash-Based Storage Systems, Big Data Processing, and Cloud Computing, Atmiya University, Rajkot, March 12, 2026.
3. Guest Speaker: Industry, Academia, and Startup: A First-Hand Innovation Journey, BITS Pilani, February 17, 2026.
4. Guest Speaker: Advancing System Design: Classical and Quantum, BITS Pilani, February 4, 2026.
5. Guest Speaker: A Comprehensive Approach to Memory and Storage Systems Optimization, NSF AI Institute, Online, February 28, 2024.
6. Guest Speaker: Picking Research as Career, Women in CS (WiCS) Student Chapter, Miami, FL, April 17, 2023.
7. Guest Speaker: Research Towards Data Storage and Management, Presentation Request for Flit-Path Scholars, Miami, FL, February 11, 2022.
8. Invited Speaker: Emerging Technologies Moving Forward, Entrepreneurs' Organization (EO), Miami, FL, February 10, 2022.
9. Guest Lecture: Towards Designing Intelligent Storage Devices, IBM Research, Almaden, San Jose, CA, May 12, 2021.
10. Guest Lecture: Challenges of the Evolving Memory and Storage Technologies, Memory Solutions Lab, Samsung, San Jose, CA, October 23, 2020.
11. Guest Speaker: Ph.D. in Computer Science – from the lens of a Girl who likes pink, FIU Women in Cybersecurity (WiCys) Student Chapter, Miami, FL, October 22, 2020.
12. Guest Lecture: New Techniques for Data Management in Evolving Storage Technologies, Florida International University, Miami, FL, November 22, 2019.
13. Guest Lecture: New Storage Technologies for Big Data Processing on Cloud and Datacenter Infrastructures, Colorado State University, Fort Collins, CO, March 27, 2019.
14. Paper Presentation Talk: FIOS: Feature Based I/O Stream Identification for Improving Endurance of Multi-Stream SSDs, 2018 IEEE International Conference on Cloud Computing (CLOUD'18), San Francisco, CA, 2018.
15. Paper Presentation Talk: BloomStream: Data Temperature Identification for Flash Based Memory Storage Using Bloom Filters, 2018 IEEE International Conference on Cloud Computing (CLOUD'18), San Francisco, CA, 2018.
16. Paper Presentation Talk: FiM: Performance Prediction Model for Parallel Computation in Iterative Data Processing Applications, IEEE International Conference on Cloud Computing (CLOUD'17), Honolulu, HI, 2017.
17. Paper Presentation Talk: Understanding Performance of I/O Intensive Containerized Applications for NVMe SSDs, IEEE International Performance Computing and Communications Conference (IPCCC'16), Las Vegas, NV, 2016.
18. Paper Presentation Talk: Accelerating Big Data Applications Using Lightweight Virtualization Framework on Enterprise Cloud, IEEE High Performance Extreme Computing Conference (HPEC'17), Waltham, MA, 2017.
19. Paper Presentation Talk: Design Space Exploration of GPU Accelerated Cluster Systems for Optimal Data Transfer Using PCIe Bus, IEEE High Performance Extreme Computing Conference (HPEC'16), Waltham, MA, 2016.
20. Paper Presentation Talk: Accelerating K-Means Clustering with Parallel Implementations and GPU

Computing, IEEE High Performance Extreme Computing Conference (HPEC'15), Waltham, MA, 2015.

PATENT DISCLOSURES, APPLICATIONS, AND AWARDS

(Content in blue color are items since arriving at FIU.)

1. Christopher Lukas Kverne and Janki Bhimani inventors; "Quantum Neural Network Checkpointing," Application.
2. Omkar Desai, Shuyi Pei, Janki Bhimani, and Bryan Kim inventors; "Cache Systems and Methods for Training Machine Learning Models", Application.
3. Daniel Carlson*, Adnan Maruf*, Raju Rangaswami, and Janki Bhimani, inventors; "Techniques to Dynamically Allocate Pages within CXL Memory Systems", Application.
4. Manoj Pravakar Saha*, Yanzhao Wu, Raju Rangaswami, and Janki Bhimani, inventors; "Systems and Methods for Checkpointing DNN", Application.
5. Manoj Pravakar Saha*, Janki Bhimani, inventors; "Systems and methods for optimizing data management within key value storage", US Patent 11,474,699.
6. Adnan Maruf*, Ashikee Ghosh*, Raju Rangaswami, and Janki Bhimani, inventors; "ML based Tiered Memory", US Patent 11,307,986.
7. Janki Bhimani, Jingpei Yang, Changho Choi, inventors; Samsung Electronics Co Ltd, assignee. "Parallel key value based multi-thread machine learning exploiting KV-SSDs" US Patent 11,099,790.
8. Janki Bhimani, Rajinikanth Pandurangan, Changho Choi, Vijay Balakrishnan, inventors; Samsung Electronics Co Ltd, assignee. "System and method for identifying hot data and stream in a solid-state drive" US Patent 10,901,907.
9. Janki Bhimani, Rajinikanth Pandurangan, Vijay Balakrishnan, Changho Choi, inventors; Samsung Electronics Co Ltd, assignee. "Methods and systems for testing storage devices via a representative I/O generator" US Patent 10,956,294.
10. Janki Bhimani, Anand Subramanian, Vijay Balakrishnan, and Jingpei Yang, inventors; Samsung Electronics Co Ltd, assignee. "Container workload scheduler and methods of scheduling container workloads" US Patent 10,719,354.
11. Janki Bhimani, Jingpei Yang, Changho Choi, Jianjian Huo, inventors; Samsung Electronics Co Ltd, assignee. "Smart I/O stream detection based on multiple attributes" US Patent 10,824,576 and 10,282,324.
12. Janki Bhimani, Hingkwan Huen, Jingpei Yang, Manu Awasthi, Vijay Balakrishnan, Jason Martineau, inventors; Samsung Electronics Co Ltd, assignee. "Intelligent controller for containerized applications" US Patent 10,210,024.

ACADEMIC SUPERVISION

Doctoral Students (Thesis Advisees)

Graduated Ph.D. Students

1. Adnan Maruf, Ph.D. student
   Dissertation topic: Improving the performance and reliability of systems with emerging memory and storage devices
   Graduated in Apr. 2023
   Tenure Track Assistant Professor, Missouri State University
2. Manoj Pravakar Saha, Ph.D. student
   Dissertation topic: Investigating emerging data storage strategies to improve unstructured data management and deep learning performance

Graduated in Jul. 2025
Tenure Track Assistant Professor, Utah Valley University

Current Ph.D. Students (Thesis Advisees)

3. Alexis Gonzales, Ph.D. student
   Dissertation topic: Fault-Tolerant Caching Service
   Dissertation plan: Fall 2027
4. Mayur Akewar, Ph.D. student
   Dissertation topic: Adaptive Decision-Making for Operational Intelligence and Resilient Execution Across Classical and Quantum Computing Systems
   Dissertation plan: Spring 2029
5. Sri Ram Durga Prasad Sarvasiddi, Ph.D. student
   Dissertation topic: Towards Designing Intelligent Techniques for Reliable Quantum-Classical Computing
   Dissertation plan: Spring 2030
6. Orkhan Fataliyev, Ph.D. student
   Dissertation topic: Adaptive Neural Architecture Control for Resource-Constrained Model Training
   Dissertation plan: Spring 2031
7. Chandrabhan Patel, Ph.D. student
   Dissertation topic: Towards Designing New Techniques for AI Based Data Indexing and Neuro-Symbolic AI
   Dissertation plan: Spring 2032

Discontinued Ph.D. Students

8. Ali Bin Omer Qureshi, Ph.D. student
   Dissertation topic: Optimizing Data Spilling in Distributed Query Engine and Memory Management
   Duration advised by me: Fall 2023 - Summer 2025
9. Daniel Carlson, Ph.D. student
   Dissertation topic: Improving the Performance of Dis-aggregated Memory Systems
   Duration advised by me: Fall 2020 - Spring 2024
10. Ashikee Ghosh, Ph.D. student
    Dissertation topic: Designing Libraries for Efficient ML Checkpointing
    Duration co-advised by me: Spring 2020 - Fall 2023
    Software Development Engineer, Amazon
11. Maimuna Begum Kali, Ph.D. student
    Dissertation topic: Optimizing Parallel Operations within BigData Processing Platforms
    Duration advised by me: Fall 2019 - Fall 2021
    Ph.D. student, School of Universal Computing, Construction, and Engineering Education (SUCCEED)

Ph.D. Committee Member

12. John Dabruzzo
13. Sagor Chandro Bakchy
14. Trinity Socorro
15. Claire Huo (Rice University)
16. Sameer Narendra Jadhav
17. Pratik Poudel
18. Pedro Espina
19. Sumesh Kumar
20. Omkar Desai (Syracuse University)
21. Rafael Trujillo (Graduated)

22. Ziyang Jiao (Syracuse University) (Graduated)
23. Liana Valdes Rodriguez (Graduated)
24. Oswaldo Artiles (Graduated)
25. Muhammad Haseeb (Graduated)
26. Danlin Jia (Northeastern University) (Graduated)

M.S. Students (Thesis/Project Advisees)

Graduated M.S. Students

27. Dwaraka Prasath Mohen Babu, ESpace Networks
28. Ashikee Ghosh, Amazon
29. Ali Bin Omer Qureshi
30. Shashidhar Reddy Chavula
31. Muttahar Khalid

Research Experience for Undergrad (REU) Students

Graduated Undergrad Students

32. Aarush Gupta
33. Priyanshi Arvind Modi
34. Leora Modi
35. Romir Shetty
36. Omkar Shrikanth
37. Shashwat Patni
38. Parag Pal
39. Aryan Singh
40. Priyanshu Gupta
41. Christopher Lukas Kverne
42. Ana Oliveria
43. Lorena a. Quincoso Lugones
44. Adrian Gonzalez
45. Niccholas Reiz
46. Amanda Di Perna, Program Manager, Debris Free Oceans
47. Federico Monteverdi, Software Engineer, Progressive
48. Gabriel Zavala, Software Engineer, Dell
49. Daniel Carlson
50. Roberto Martinez, Co-Founder & CPO of GammaSwap Labs
51. Christopher Meadows
52. Aris Duani Rojas, Ph.D. Student
53. Sashri Brahmakshatriya
54. Natalia Valencia, Ph.D. Student
55. Kevin Nordman

Current Undergrad Students

56. Lenish Pandey
57. Anxhela Nano
58. Bianca Poggi

Independent Study

1. Olga Perdomo, Fall 2026, Topic: Quantum in fin-tech.
2. Daniel Carlson, Summer 2022, Topic: Hybrid Memory Management.
3. Dwaraka Prasath Mohen Babu, Spring 2022, Topic: Data Structures to Identify Data Streams.
4. Sashri Brahmakshatriya, Summer 2021, Topic: Analyze Reliability of SSDs.
5. Christopher Meadows, Summer 2021, Topic: Design Data Stream Identifier.

Capstone Mentoring

1. Daniel Carlson
2. Patrick Perez
3. Oscar Barbosa
4. Nazmul Huq
5. Luis Acosta
6. Ettore Mottola
7. Eitan Flor
8. Bryan Camacho

Teaching Activities

Quantum Algorithms (COT 5600): Developed and taught a graduate-level course on quantum algorithms at Florida International University, covering the theoretical foundations, design principles, and computational complexity of quantum algorithms. The course explores fundamental and advanced algorithms, including quantum search, quantum Fourier transform, phase estimation, Shor's factoring algorithm, Grover's search algorithm, quantum walks, and variational quantum algorithms. Emphasis is placed on understanding the quantum speedups offered by these algorithms, analyzing their computational complexity, and implementing and evaluating quantum algorithms using quantum programming frameworks and simulators. The course also examines current challenges and emerging applications of quantum algorithms on near-term and fault-tolerant quantum computing platforms.

Fundamentals of Quantum Computing (COT 4601): Developed and taught an introductory course on quantum computing and quantum algorithms at Florida International University, covering the foundations of quantum theory, quantum states and gates, superposition, entanglement, quantum circuits, and quantum software development. The course emphasizes hands-on exploration of fundamental quantum algorithms, including Grover's search and Shor's factoring algorithms, through simulation and practical implementation. Students also examine quantum error correction, quantum hardware limitations, and emerging applications of quantum computing.

Storage Systems (CIS 5346): In my most recent offering of this course, it underwent evaluation by students through the Student Perceptions of Teaching Survey (SPOTs), yielding a commendable mean score of 4.64/5. Notable comments from students include, *"The most successful aspect of this course is the detailed information in the lectures about the objectives and executions of each module and great instructor-student interactions."* Another student mentioned, *"The professor's availability and commitment towards every student was commendable."* Additionally, a student highlighted, *"This was the best online course that I have taken so far, expertly crafted, and the pacing of this course was good."* Lastly, a student appreciated the practicality, stating, *"The way we could relate the course to real-life scenarios will definitely help me never forget what I learned. Course material and discussions are thought-provoking and interesting."*

Data Structures (COP 3530): COP 3530 earns its "effort-intensive" label due to students' substantial

growth expectations post-course and a high enrollment rate. My SPOTs rating was 4.59/5. Students praised COP 3530 for being "*well-structured,*" with one noting, "*The most successful aspect is probably having discussions for each Module.*" Another student commended the "*Professor's teaching proficiency is 5+ stars rating, she presents lectures in a clear, understandable way with many opportunities in each module to get clarification.*" The "video lectures" were highlighted for their "*easy-to-understand quality.*" Overall, the course was recognized for its "*balance, pacing, and real-world applicability, making it a transformative and great learning experience.*" The peer evaluation of this course by my colleague Dr. Masoud Milani stated that "*my observations, as well as students' opinions, confirm that Dr. Bhimani is an outstanding teacher who has clearly mastered the art of teaching.*"

OTHER PROFESSIONAL ACTIVITIES AND PUBLIC SERVICE

FIU Internal Service

1. 2026-2027: CEC Faculty Council Representative
2. 2026-2027: KFSCIS PhD Admissions Committee (Chair)
3. 2026-2027: KFSCIS Ad-Hoc Committee on Research Evaluation Rubrics
4. 2024-2025: KFSCIS Graduate Program Committee (GPC)
5. 2024-2025: KFSCIS Tenure and Tenure-Track Faculty Search and Screen Committee (TTT Hiring)
6. 2024-2025: CEC Faculty Council Representatives
7. 2023-2024: KFSCIS Awards Committee
8. 2023-2024: CEC Faculty Council Representative
9. 2023-2024: KFSCIS Seminar Series Coordinator
10. 2023-2024: CEC Faculty representative at United Nations Women and Girls in Science day annual celebrations
11. 2023-2024: Graduate Council
12. 2022-2023: CEC Faculty Council Representative
13. 2022-2023: KFSCIS Seminar Series Coordinator
14. 2022-2023: Subject area coordinator: BS-CS for Programming: COP-2210, COP-3337, COP-3530, COP-4338, COP-4226, COP-4520
15. 2022-2023: KFSCIS Diversity, Equity & Inclusion (DEI) Committee
16. 2022-2023: CEC Faculty representative at United Nations Women and Girls in Science day annual celebrations
17. 2022-2023: Graduate Council
18. 2021-2022: KFSCIS Tenure Track Faculty Hiring Committee
19. 2021-2022: Subject area coordinator: BS-CS for Programming: COP-2210, COP-3337, COP-3530, COP-4338, COP-4226, COP-4520
20. 2021-2022: Capstone or Senior Projects Supervisor
21. 2021-2022: KFSCIS Diversity Advocate for Faculty Hiring
22. 2021-2022: CEC Faculty representative at United Nations Women and Girls in Science day annual celebrations
23. 2021-2022: Graduate Council
24. 2020-2021: KFSCIS Tenure Track Faculty Hiring Committee
25. 2020-2021: Subject area coordinator: BS-CS for Programming: COP-2210, COP-3337, COP-3530, COP-4338, COP-4226, COP-4520

26. 2020-2021: Capstone or Senior Projects Supervisor
27. 2020-2021: KFSCIS Diversity Advocate for Faculty Hiring
28. 2020-2021: KFSCIS Faculty representative at United Nations Women and Girls in Science day annual celebrations
29. 2020-2021: Graduate Council
30. 2020-2021: KFSCIS Graduate Committee
31. 2019-2020: CEC Faculty representative at United Nations Women and Girls in Science day annual celebrations
32. 2019-2020: Graduate Council
33. 2019-2020: KFSCIS Graduate Committee

FIU Microcredential

Remote Teach Ready Badge Summer 2020

Professional Activities

1. 2026 NSF Panelist
2. 2026 Associate Editor for ACM Transactions on Architecture and Code Optimization (ACM TACO).
3. 2026 TPC for USENIX Conference on File and Storage Technologies (USENIX FAST) 2025 with heavy review workload of 15-20 papers in each round and two rounds of submission.
4. 2026 PC member for ACM Workshop on Hot Topics in Storage and File Systems (HotStorage).
5. 2025 Associate Editor for ACM Transactions on Architecture and Code Optimization (ACM TACO).
6. 2025 NSF Panelist.
7. 2025 TPC for USENIX Conference on File and Storage Technologies (USENIX FAST) 2025 with heavy review workload of 15-20 papers.
8. 2024 Associate Editor for ACM Transactions on Architecture and Code Optimization (ACM TACO).
9. 2024 Panelist on a Panel to discuss ML for Storage at HotStorage'24.
10. 2024 General Chair for ACM Workshop on Hot Topics in Storage and File Systems (HotStorage) 2024 leading the overall operations of the workshop.
11. 2024 TPC for USENIX Conference on File and Storage Technologies (USENIX FAST) 2024 with heavy review workload of 15-20 papers.
12. 2024 Program Committee Track chair for the track Performance Monitoring, Modeling, Analysis, and Benchmarking (in Cluster, Cloud and Internet Computing) at 22nd IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid).
13. 2024 Poster chair for the 33nd ACM International Symposium on High-Performance Parallel and Distributed Computing (HPDC).
14. 2023 Poster chair for the 32nd ACM International Symposium on High-Performance Parallel and Distributed Computing (HPDC).
15. 2023 Publicity chair and TPC for ACM Workshop on Hot Topics in Storage and File Systems (HotStorage).
16. 2022 TPC and Session chair for ACM Workshop on Hot Topics in Storage and File Systems (HotStorage) 2022 leading a session on ZNS and SSDs.
17. TPC for IEEE International Conference on CLOUD Computing
18. 2022 TPC for IEEE International Conference on Distributed Computing Systems (ICDCS), Machine Learning on or for Distributed Systems
19. 2022 TPC for IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing

(CCGrid), Performance Modeling, Scheduling and Analysis Track

20. 2023 NSF Panelist
21. 2022 NSF Panelist
22. 2021 TPC and Session chair for USENIX Conference on File and Storage Technologies (USENIX FAST), leading a session on The SSD Revolution Is Not Over.
23. 2021 TPC and Session chair for ACM Workshop on Hot Topics in Storage and File Systems (HotStorage) leading a session on Flash Storage, 2021.
24. 2020 TPC and Session chair for IEEE International Symposium on Workload Characterization (IISWC), leading a session on System Architecture and Applications.
25. 2020 TPC for IEEE International Parallel & Distributed Processing Symposium (IPDPS).
26. 2020 TPC for IEEE International Performance Computing and Communications Conference (IPCCC).
27. 2019 TPC for IEEE International Performance Computing and Communications Conference (IPCCC).

Service as Peer Reviewing

Conferences:

1. Conference on Neural Information Processing Systems (NeurIPS)
2. International Conference on Learning Representations (ICLR)
3. International Conference on Machine Learning (ICML)
4. Peer Reviewer: AAAI Conference on Artificial Intelligence (AAAI)
5. IEEE International Conference on Distributed Computing Systems (ICDCS)
6. IEEE/ACM International Symposium on Cluster, Cloud, and Internet Computing (CCGRID)
7. ACM Workshop on Hot Topics in Storage and File Systems (HotStorage)
8. International Symposium on High-Performance Parallel and Distributed Computing (HPDC)
9. USENIX Conference on File and Storage Technologies (FAST)
10. IEEE International Parallel & Distributed Processing Symposium (IPDPS)
11. IEEE International Conference on Cloud Computing (IEEE CLOUD)
12. IEEE High Performance Extreme Computing Conference (IEEE HPEC)
13. IEEE International Conference on Green Computing and Communications (GreenCom)
14. International Conference on Massive Storage Systems and Technology (MSST)
15. IEEE International Conference on Big Data (BigData)
16. International Conference on Networking, Architecture, and Storage (NAS)
17. International Conference on Parallel and Distributed Systems (ICPADS)
18. Workshop on Interactions of NVM/Flash with Operating Systems and Workloads (INFLOW)
19. International Conference on Performance Engineering (ICPE)
20. ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA)
21. IEEE/IFIP International Conference on Dependable Systems and Networks (DNS)
22. Big Data and Cloud Performance Workshop at INFOCOM (DCPerf)
23. International Conference on Autonomic Computing (ICAC)
24. International Conference on Computer Aided Design (ICCAD)
25. International Conference on Cloud Computing Technologies and Applications (CloudTech)
26. Field-Programmable Custom Computing Machines (FCCM)
27. International Conference on Computer. Communication and Networks (ICCCN)

28. IEEE International Performance Computing and Communications Conference (IPCCC)
29. IEEE/ACM International Conference on Utility and Cloud Computing (UCC)

Journals:

1. IEEE Access
2. ACM Transactions on Architecture and Code Optimization (TACO)
3. ACM Transactions on Embedded Computing Systems (TECS), ACM Journal
4. IEEE Transactions on Cloud Computing (TCC), IEEE Journal
5. ACM Transaction on Storage (TOS), ACM Journal
6. IEEE Transactions on Services Computing (TSC), IEEE Journal
7. Simulation Modelling Practice and Theory (SIMPAT), Elsevier Journal
8. Computers, MDPI Journal
9. Future Generation Computer Systems (FGCS), Elsevier Journal
10. IEEE Transactions on Computers (TC), IEEE Journal
11. ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS), ACM Journal

Books:

1. CRC Press-Taylor & Francis- Book Proposal: Introduction to Data Storage Systems and Services

Society Memberships

1. Member (2025-present) American Society for Engineering Education (ASEE)
2. Member (2014-present) Association for Computing Machinery (ACM)
3. Member (2014-present) Institute of the Electrical and Electronics Engineers (IEEE)
4. Member (2014-present) The Advanced Computing Systems Association (USENIX)

Community Services

1. Volunteering Affiliated Faculty, Center for Women and Gender Studies (CWGS), Florida International University, Miami FL, USA

CREATIVE WORK

1. Mentored new Assistant Professors in navigating and excelling in the multifaceted responsibilities of faculty life, including grant writing, research development, Ph.D. student advising and mentorship, teaching, and academic service, providing practical guidance and mentorship in areas that are often not explicitly covered during doctoral training.
2. Developed a faculty-student research engagement platform to streamline the process of identifying, communicating, and matching undergraduate students with research opportunities, increasing accessibility to research experiences and reducing administrative overhead.
3. Developed a departmental rubric for evaluating faculty research progress, establishing a transparent and consistent framework for assessing research productivity, impact, and professional growth.
4. Led the development and approval of the NSF Broadening Participation in Computing (BPC) Plan for the KFCISE department, providing a standardized NSF approved document for faculty submitting medium and large grants, including CORE and CAREER proposals.
5. Integrated class projects with the NSF Chameleon Cloud to prepare students for utilizing shared cloud computing platforms in their research, enhancing their skills in cloud infrastructure, scalability, and resource management.

6. Conducting a SWOT Analysis as a strategic planning technique has been a key initiative to enhance the overall learning experience of the class.
7. Taking the lead in designing and developing the website for the VOCES project, which can be accessed at https://voces.fiu.edu/, has been a significant contribution.
8. Creating an efficient workflow that covers the entire lifecycle of multimedia posts, including stages such as concept brainstorming, assignment of responsibilities, post design using tools like Canva, feedback collection and review, post enhancements, and final staging, has streamlined the process.
9. Determining the optimal posting frequency and content distribution across various social media platforms, such as LinkedIn, Instagram, and others, with the aim of organically boosting engagement and cultivating student interest in becoming VOCEROs, has been a strategic focus.
10. Creating a pioneering teaching method by incorporating surveys and polls to assess student engagement and pinpoint challenging topics has been a transformative approach. Initially breaking down module topics into more digestible sub-topics, prompting students to assess their difficulty through pre-class polls, and then revisiting the sub-topics that most students found challenging in the following class using real-time feedback, employing personalized teaching techniques to enhance understanding, and facilitating an open forum for students to anonymously ask questions has significantly improved the learning experience.